Profit P = Total Revenue – Total cost

$$= 60x - 50x - \frac{x^2}{100} - 200$$

$$= 10x - \frac{x^2}{100} - 200$$

$$\frac{dp}{dx} = 10 - \frac{x}{50}$$

To get profit maximum, $\frac{dp}{dx} = 0 \Rightarrow \boxed{x = 500.}$

$$\frac{d^2P}{dx^2} = \frac{-1}{50} < 0$$

$\therefore$ Profit is maximum when $x = 500$ and

Maximum Profit is $P = 10(500) - \frac{(500)^2}{100} - 200$

$= 5000 - 2500 - 200$

$= 2300$

Profit = 2,300.

Example 3.19

The marginal cost and marginal revenue with respect to commodity of a firm are given by $C'(x) = 8 + 6x$ and $R'(x) = 24$. Find the total Profit given that the total cost at zero output is zero.

Solution:

Given $MC = 8 + 6x$

$$C(x) = \int (8 + 6x)\,dx + k_1$$

$$= 8x + 3x^2 + k_1 \quad (1)$$

But given when $x = 0, C = 0 \Rightarrow k_1 = 0$

$\therefore$ $C(x) = 8x + 3x^2$ (2)

Given that $MR = 24$

$$R(x) = \int MR\,dx + k_2$$

$$= \int 24\,dx + k_2$$

$$= 24x + k_2$$

Revenue = 0, when $x = 0 \Rightarrow k_2 = 0$

$R(x) = 24x$ (3)

Total Profit functions $P(x) = R(x) - C(x)$

$$P(x) = 24x - 8x - 3x^2$$

$$= 16x - 3x^2$$

Example 3.20

The marginal revenue function (in thousand of rupees) of a commodity is $10 + e^{-0.05x}$ Where x is the number of units sold. Find the total revenue from the sale of 100 units $(e^{-5} = 0.0067)$.

Solution:

Given, Marginal revenue $R'(x) = 10 + e^{-0.05x}$

Total revenue from sale of 100 units is

$$R = \int_0^{100} \left(10 + e^{-0.05x}\right) dx$$

$$= \left[10x + \frac{e^{-0.05x}}{-0.05}\right]_0^{100}$$

$$= \left(1000 - \frac{e^{-5}}{0.05}\right) - \left(0 - \frac{100}{5}\right)$$

$$= 1000 + 20 - (20 \times 0.0067)$$

$$= 1019.87$$

Total revenue = 1019.87×1000

= ₹10,19,870

Example 3.21

The price of a machine is ₹5,00,000 with an estimated life of 12 years. The estimated salvage value is ₹30,000. The machine can be rented at ₹72,000 per year. The present value of the rental payment is calculated at 9% interest rate. Find out whether it is advisable to rent the machine. $(e^{-1.08} = 0.3396)$.

Solution:

The present value of payment for t year = $\int_0^t 72000\,e^{-0.09t}\,dt$

Present value of payment for12 years = $\int_0^{12} 72000\,e^{-0.09t}\,dt$

$$= 72000\left[\frac{e^{-0.09t}}{-0.09}\right]_0^{12}$$

$$= \frac{72000}{-0.09}\left[e^{-0.09(12)} - e^0\right]$$

$$= -8,00,000\left[e^{-1.08} - e^0\right]$$

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